

Evaluation of in-vehicle HMI using occlusion techniques: experimental results and practical implications

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Abstract

Despite the usefulness of new on-board information systems one has to be concerned about the potential distraction effects that they impose on the driver. Therefore, methods and procedures are necessary to assess the visual demand that is connected to the usage of an on-board system. The occlusion-method is considered a strong candidate as a procedure for evaluating display designs with regard to their visual demand. This paper reports results from two experimental studies conducted to further evaluate this method. In the first study, performance in using an in-car navigation system was measured under three conditions: static (parking lot), occlusion (shutter glasses), and driving. The results show that the occlusion-procedure can be used to simulate visual requirements of real traffic conditions. In a second study the occlusion method was compared to a global evaluation criterion based on the total task time. It can be demonstrated that the occlusion method can identify tasks which meet this criterion, but are yet irresolvable under driving conditions. It is concluded that the occlusion technique seems to be a reliable and valid method for evaluating visual and dialogue aspects of in-car information systems.

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Keywords: Occlusion-technique; Visual attention; Information systems

1. Introduction

Driving is to a high degree a visual task. Most of the information perceived by a driver is based on the visual channel and impairment of visual information processing is considered a main reason for accidents. Therefore everything that possibly distracts people from the primary driving task with their eyes-on-the-road and their hands-on-the-wheel has to be investigated with regard to safety considerations. These considerations include (among other things) that the information presented to a driver can be interpreted within a tolerable time scale (see Zwahlen et al., 1988) and that dialogue sequences between a driver and an in-car information system can easily be interrupted and resumed in order to minimise distraction (see Antin, 1993). In order to prove that these requirements are met by the system a procedure is necessary to evaluate different HMI concepts. The occlusion method is

proposed to be such a procedure. Although occlusion during driving was used in several studies to determine the visual demand of different traffic situations (see van der Horst, in this volume, for an overview), the approach presented in this paper is different. Here the technique is used for simulating driving conditions where a driver has to shift visual attention between the car periphery situation and an on-board system. In our previous research we could show that the occlusion technique used in this way might be a valid and easy to use procedure for assessing the visual demand of a task. One major requirement for display-based on-board systems like navigation systems is that information must be readable and understandable fast and reliably. If the occlusion technique can be used to test whether a system fulfils this requirement it should discriminate between displays which are associated with different processing demands. This was investigated in a previous series of experiments (see Krems et al., 2000). We found that the occlusion technique discriminated reliably between two different versions of displays (complex versus simple) in processing time and error probability. A second series of experiments addressed the issue of

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dialogue interruption (see Keinath et al., 2001). In a visual search task it could be shown that chunkability of dialogs might be assessed by the occlusion technique.

The major aim of this paper is to report results from two experimental studies where the validity and usability of the occlusion technique as a method for evaluating visual aspects of in-car information systems was further investigated. First, results of a field study conducted to assess the validity of the method outside the laboratory will be presented. Second, findings from a study comparing the occlusion technique with an alternative method that uses global measures like total task time to evaluate in-car information systems will be given.

2. Experiment 1: comparison of performance under occlusion and traffic conditions

The results of the laboratory experiments so far suggest that the occlusion technique is a useful procedure for evaluating displays. What is still missing is the validation of the technique outside the laboratory. Therefore, an experiment was conducted to determine how well the performance in in-car tasks under occlusion conditions compares to the performance under driving conditions. The major aim of this study was to investigate how the various measures, like error rates or inspection times, obtained through evaluation of a system by means of the occlusion technique relate to the same measures obtained while driving a car and simultaneously interacting with an on-board information system. The same tasks (entering a destination in a navigation system) and the same material (address charts) were used under different conditions.

2.1. Method

Participants: 43 participants took part in this study (age $M = 23.4$, $SD = 4.1$; 11 male, 32 female) for course credit.

Material: As stimuli we used 15 address charts indicating the destination to be entered into the navigation system. An address consisted of a city name and a street name. One-third of the destinations involved 5–8 inputs to complete the task (short input), one-third 9 to 11 (medium input) and one-third 12 to 13 (long input).

Task and Equipment: The participants' task was to enter each of the 15 destinations into the BMW Carin navigation system. It is operated by rotating and pressing one single button next to the display in the console. Letters are selected by rotating the button until the target letter is reached and then pressing the button. The system compares the given input with a database of predefined city and street names. If a match is found the input is completed by the system as fully as possible.

Under the *non-occlusion condition* participants had to perform the task in a parked car without any distraction. Under the *occlusion condition* participants had to wear shutter glasses (PLATO LC Shuttering Spectacles, Milgram, 1987) while entering the destinations. The inspection times were 1.5 s, which was within the maximum time tolerance range for taking one's eyes off the road (Zwahlen et al., 1988). The occlusion time was 3 s. Under the *driving condition* participants had to enter the addresses into the navigation system while driving a car. The route under the driving condition was selected to be on the outskirts of a city.

Procedure: The participants took place in the driver's seat. They were then trained on the handling of the navigation system and entered two destinations for practice. For the driving condition, participants were also given some instruction and practice-time regarding driving (BMW sedan, automatic transmission) to allow them to familiarise themselves with the car before the experiment began. At the beginning of each of the 15 trials the experimenter—seated in the front passenger seat—gave each participant a chart indicating a destination to be entered. The participant then had to pronounce the destination and thereafter to enter the destination into the system. The destinations were presented randomly. All reactions of the participants were videotaped.

Design: A two-factorial mixed design was used in this experiment. The input condition (non-occlusion, occlusion, and driving) was varied as a between-subjects factor. There were 13 participants under the driving condition, 16 under the non-occlusion condition, and 14 under the occlusion condition. Participants were randomly assigned to the three conditions.

The second factor, length of input, was varied as a within-subjects factor with three levels, short, medium, and long input with five examples each. The amount of errors and the time to complete each trial were measured as dependent variables.

2.2. Results

Because of missing data one participant of the traffic condition, three of the non-occlusion condition, and two of the occlusion condition could not be included in the analysis of total task times and errors. Three variables were defined according to the input condition: the total task time for solving the task without driving and without occlusion under static conditions ($TTT_{\text{non-occl}}$), the total task time under occlusion condition (TTT_{occl}), and the total time to solve the task while driving (TTT_{driving}). Fig. 1 shows the total task times as a function of input condition and input length.

More time was needed under the driving condition ($TTT_{\text{driving}} = 110.3$ s, $SD = 39.8$ s) to complete a task than under the occlusion condition ($TTT_{\text{occl}} = 93.8$ s,

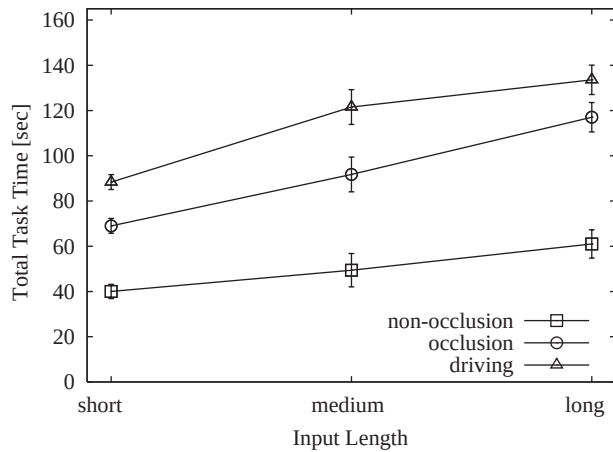


Fig. 1. Total task times as a function of input condition and input length.

SD=24.7 s) or under the non-occlusion condition ($TTT_{\text{non-occl}} = 50.4$ s, SD=10.6 s). As expected the time to complete a task increased with the length of the input ($M_{\text{short}} = 63.7$ s, $SD_{\text{short}} = 22.7$ s, $M_{\text{medium}} = 83.9$ s, $SD_{\text{medium}} = 38.1$ s, $M_{\text{long}} = 100.4$ s, $SD_{\text{long}} = 39.1$ s). A repeated measures two-factorial ANOVA with input condition as between-subjects factor and input length as within-subjects factor revealed significant main effects of the input condition factor, $F(2, 34) = 44.84$, $p < 0.001$, and the input length factor, $F(2, 68) = 59.78$, $p < 0.001$. Post hoc comparisons with Bonferroni correction between each pair of input conditions showed that total task times differed significantly from each other for each input condition. The same analysis for each pair of input length yielded significant differences between all pairs of input length levels. The interaction between input condition and input length was also significant, $F(4, 68) = 3.92$, $p = 0.006$. This is due to the fact that total task times increased less for long inputs under the driving condition than under both the occlusion and the non-occlusion conditions.

In order to further investigate the interrelationship between the different input conditions (non-occlusion, occlusion, driving) TTT was calculated for every task by averaging over participants within the three input conditions. The correlations between the different conditions are shown in Table 1. All correlations are highly significant; the correlation between TTT_{occl} and TTT_{driving} is slightly higher than between $TTT_{\text{non-occl}}$ and TTT_{driving} .

The total shutter open time (TSOT) is defined by the sum of all durations of the open/close cycle where the goggles were open. On average over all tasks $TSOT = 31.9$ s (SD=11.2 s). In order to estimate the portion of TTT that is combined with visual attention to the task, the ratio $TSOT/TTT_{\text{non-occl}}$ was calculated. This ratio is 0.63 for the tasks used. It can be viewed as a rough measure of the degree to which a single task can

Table 1

Correlations between total shutter open time and total task times under the three different input conditions

	Non-occlusion	Occlusion	Driving	TSOT
Non-occlusion	—	0.96 ^a	0.89 ^a	0.92 ^a
Occlusion	0.96 ^a	—	0.91 ^a	0.98 ^a
Driving	0.98 ^a	0.91 ^a	—	0.90 ^a
TSOT	0.91 ^a	0.98 ^a	0.90 ^a	—

Note. ^a $p < 0.01$.

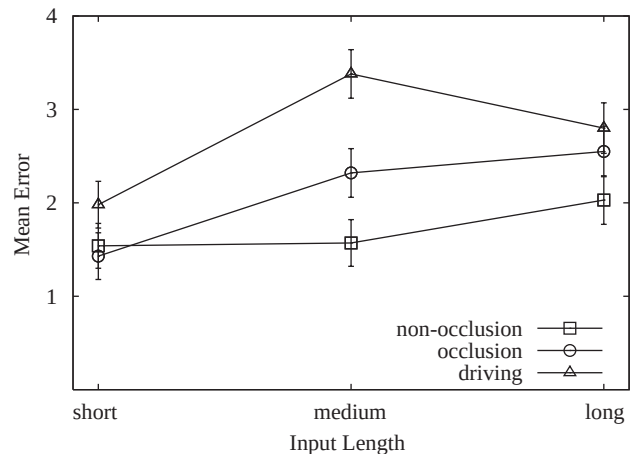


Fig. 2. Mean number of errors as a function of input condition and input length.

be done without visual control. The smaller the deviation from 1 the more visual attention would be necessary for performing a task.

For the error analysis all trials where the number of errors exceeded the critical value of the median of each condition plus three times the interquartile range of the condition, were treated as outliers and were excluded from data analysis.

Fig. 2 shows the average number of errors per trial and participant as a function of the input condition and length. The fewest errors were made under the non-occlusion condition ($M = 1.7$, $SD = 1.6$). Under the occlusion condition an average of 2.1 errors were made ($SD = 1.9$), and most errors were observed under the driving condition ($M = 2.7$, $SD = 2.1$). As expected the number of errors increased with the length of the entry ($M_{\text{short}} = 1.6$, $SD_{\text{short}} = 0.9$; $M_{\text{medium}} = 2.4$, $SD_{\text{medium}} = 1.1$; $M_{\text{long}} = 2.5$, $SD_{\text{long}} = 0.97$).

A two-factorial ANOVA with input condition and length as factors revealed significant main effects of the input condition factor, $F(2, 34) = 7.13$, $p = 0.003$, and input length factor, $F(2, 68) = 14.3$, $p < 0.001$. Scheffé Post hoc tests showed that there was a significant difference between the non-occlusion and the driving condition ($p = 0.003$) and a marginally significant difference between the occlusion condition and the

driving condition ($p = 0.093$). The significant main effect of the input length factor was due to a significant difference in error rates between short and all other inputs ($p < 0.001$). Medium and long inputs did not differ from each other. The interaction between the input condition and input length also reached significance, $F(4, 68) = 3.36$, $p = 0.014$. This seems to be the case because of the unexpected drop in error rates for long inputs under the traffic condition.

In order to get a more detailed picture of the influence of input condition and input length on the errors participants made, all errors were classified into five categories. These categories reflect different levels of error severity. Their definitions together with the total number of errors in each category are shown in Table 2. Categories 1 and 2 define errors where the cursor is moved to a wrong position. Errors of this kind are caused by a common strategy participants used for moving the cursor to the intended position. This strategy consists of a first rough movement towards the intended position followed by corrective actions and fine tuning. The other three error categories define errors where a wrong command is invoked, for example entering a wrong letter or invoking a wrong menu. After such an error occurred, error recovery strategies are necessary. As errors in categories 3–5 are quite distinct from errors in categories 1 and 2, and as errors in categories 1 and 2 occur much more often than errors in the other categories only errors in category 1 and 2 were analysed statistically.

For errors in category 1, an ANOVA revealed a significant main effect of input length, $F(2, 68) = 17.6$, $p < 0.01$, a marginally significant main effect of input condition, $F(2, 34) = 2.8$, $p = 0.07$, and a significant interaction between input condition and input length, $F(4, 68) = 3.1$, $p = 0.02$. For the non-occlusion condition the average error rate was 1.4 ($SD = 1.5$), for the occlusion condition the mean error rate was 1.6 ($SD = 1.4$) and for the driving condition it was 1.8 ($SD = 1.3$). The errors averaged across input length were found to differ in a marginally significant manner for the non-occlusion condition and the driving condition ($p = 0.083$). Fig. 3 illustrates the results for error category 1.

For error category 2 (see Fig. 4), an ANOVA revealed only a marginally significant influence of the input

condition on the error rate, $F(2, 34) = 2.7$, $p = 0.07$. Again only for the overall errors for the non-occlusion condition ($M = 0.15$, $SD = 0.4$) and the driving condition ($M = 0.3$, $SD = 0.6$) were differences found to be marginally significant ($p = 0.07$).

2.3. Discussion

The participants were able to perform the task fastest and most reliably when there was no distraction or

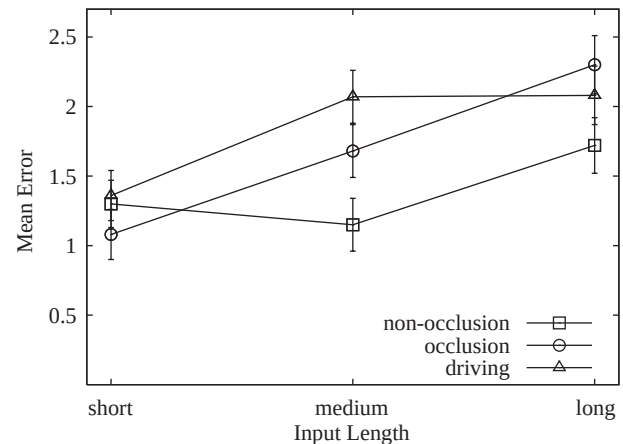


Fig. 3. Mean number of errors in error category 1 as a function of input condition and input length.

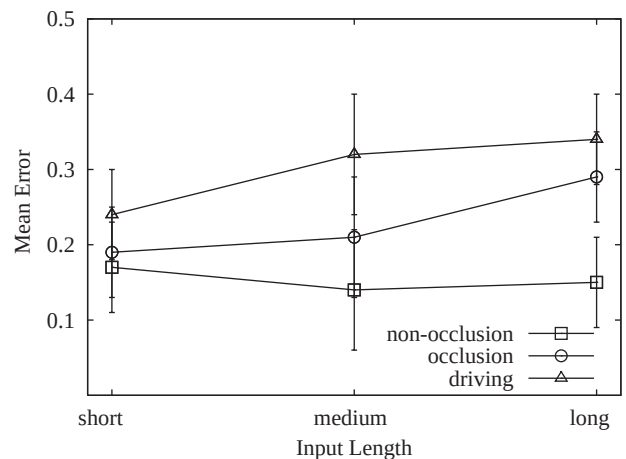


Fig. 4. Mean number of errors in error category 2 as a function of input condition and input length.

Table 2

Definition of errors and absolute number of errors in each error category

Category	Definition	Number of occurrences
Error 1	The input device was overwound for one or two positions	841
Error 2	The input device was overwound for more than two positions	111
Error 3	The participant evoked the wrong menu	54
Error 4	The participant entered the wrong letter	94
Error 5	The participant switched to the false menu level	21

interruption. If they had to perform the task and simultaneously drive a car, it took them considerably longer to finish the task and they made more errors. Participants under the occlusion condition showed an intermediate performance. It took them less time to complete the task than participants under the driving condition, but the average total task time was still significantly longer than under the non-occlusion condition. The same holds true for the error rates. Under the occlusion condition participants made fewer errors than under the driving condition, but they made more errors than participants of the non-occlusion condition.

A more detailed analysis of errors showed that this statement has to be qualified. First, the length of the input, in terms of the number of single actions necessary to complete the task, modified the effect of input condition. A clear differentiation between the three input conditions was found only for intermediate input lengths. This was due to an unexpected decrease of the amount of errors for long inputs under the driving condition. The possible cause of this decrease was the increased probability of finishing longer inputs under the driving condition while standing at traffic lights or in a traffic jam. Consequently, input of long addresses may be specifically facilitated under this condition. Secondly, if one takes a closer look at the different kinds of errors made during the interaction with the navigation system, only the non-occlusion and the driving conditions showed a marginally significant difference for both error categories. For the same error categories occlusion and driving conditions did not differ statistically.

To compare the overall performance under the three input conditions a scaling factor was computed. For total task time (TTT) this yields a factor of 1.82 between the occlusion and non-occlusion conditions ($TTT_{occl}/TTT_{non-occl}$), a factor of 2.19 between driving and non-occlusion ($TTT_{driving}/TTT_{non-occl}$), and a factor of 1.27 between the driving and occlusion conditions ($TTT_{driving}/TTT_{occl}$), that is the total task time under the driving condition was 1.27 times longer than under the occlusion condition. The same can be computed for the error rates. Here the scaling factor between the occlusion and the non-occlusion conditions amounts to 1.31, between the driving and occlusion condition to 1.29, and between the driving and the non-occlusion condition to 1.7. This means with regard to the time needed to complete a task and errors made occlusion and driving condition are much more similar compared to non-occlusion and driving condition. Of course, the scaling factors are highly specific for the parameters used in this experiment and cannot be easily generalised to other settings. But they might even result in a better fit between driving and occlusion if the occlusion parameters are properly tuned.

The close relationship between the performance in the tasks under the different input conditions is reflected by

the extremely high correlations between $TTT_{non-occl}$, TTT_{occl} , and $TTT_{driving}$. Especially the high correlation between $TTT_{non-occl}$ and $TTT_{driving}$ seems to argue for the use of measures obtained under non-occlusion conditions like total task time or total glance time as criteria for system evaluation. But at least two arguments make this conclusion premature. First, a correlation does not reflect the size of the absolute difference between two variables. The correlation would be the same whether $TTT_{driving}$ is two times or five times longer than $TTT_{non-occl}$. Secondly, $TTT_{non-occl}$ alone does not measure whether a task can be easily interrupted while driving. But this is an important dimension that has to be considered for the evaluation of in-car tasks because the driver must always be able to interrupt the task to ensure safe driving. If the task cannot be interrupted easily, either task performance is unacceptably low while driving or the driving performance is impaired in an unacceptable manner. The aim of the second experiment in this paper is to support this second argument empirically.

3. Experiment 2: total task time as a criterion for system evaluation

The second experiment raised the question whether a global measure of task performance, like the total task time or total glance time, can be used for evaluating in-car tasks as compared to the occlusion technique as a criterion for system evaluation. An example of such a global measure is the “15-second rule” (Green, 1999). According to this rule an in-car task should be feasible while driving if it takes no more than 15 s to successfully complete the task when executed without driving. How the task goal is achieved, especially how it is chunked or partitioned, is not considered. However, this is essential for evaluating whether the task can be performed while driving or not. Zwahlen et al. (1988) showed that for safety reasons drivers should not be distracted from their driving task for longer than two seconds. Therefore, it is decisive for a task and its integration in an in-car information system that the task does not require the driver’s attention for more than 1–2 s per interaction (single interaction time). That is, the driver must be able to interrupt the interaction with the in-car information system at least after 1–2 s and s/he must be able to resume the interaction after the interruption smoothly without loss in performance quality. Using the occlusion technique it is possible to directly test, whether the task can be performed successfully if its execution is interrupted after every 1–2 s.

Tijerina et al. (2000) assume total task time and single interaction time to be fairly independent. One can, for example, imagine tasks that can be finished within 15 s requiring single interaction times of less than 2 s. For

instance searching for a number in an electronic telephone book. On the other hand tasks can be designed to follow the 15-second rule but may require a single interaction time of more than 2 s for some steps, for example, searching for a certain location on a complex and rotating electronic map to decide which street to take at the next crossing. The aim of our study was to show empirically that tasks which meet a global criterion of total task time or total glance time being less than a predefined value, like the 15-seconds rule, do not necessarily meet the criterion of the single interaction time being less than 2 s. We used a dynamic task, running text in a small window resembling Radio Data System (RDS) messages, which can be found in many telematic systems. We hypothesised that even if people can perform the task within 15 s, i.e. read the displayed messages, they could not do this under occlusion conditions, where the display is regularly occluded to simulate the driver's allocation of attention while driving.

3.1. Method

Participants: 26 students (25 female, age $M=23.0$, $SD=4.2$) of Chemnitz University of Technology participated in this experiment for course credit.

Material: As stimuli we used short texts, composed of two short sentences. Each text was at maximum 40 words long. In each text two abbreviations of German motorways were mentioned. Up to three words were presented simultaneously on the screen. The presentation of a text frame lasted for 0.7 s and each text was divided into 22 frames. The text was presented sequentially without a pause between the presentation of single frames. The position of the two frames of text containing a name of a motorway was balanced across the experimental sessions. Each text was divided into four parts, two parts with a length of five frames, and two parts with a length of six frames. The position of the motorway information was balanced across these four parts. The mean duration of a trial was around 15 s.

Equipment: The experiment was run on a Psion 5mx pro handheld computer which is also suitable for in-car use. The stimuli were presented by a java-applet on a java-compatible web-browser. As in Experiment 1 we used shutter glasses under the occlusion condition to control the time the display was visible and not visible to the participants. The shutter glasses were controlled by a computer program that opened the glasses for 1.5 s and shut them for 3.5 s. Under the non-occlusion condition the participants were uninterrupted.

Task and procedure: The task of the participants was to read the presented text, to understand its meaning and to memorise the two names of the motorways by the end of each trial. At the end of each trial participants

had to tell the experimenter the names of the two motorways first and then had to answer two questions regarding the content of the text presented in the current trial. Participants had to work through a total of 30 trials. The first five trials were used as training trials to explain the procedure of the experiment and were excluded from data analysis.

Design: A between-subjects design with two experimental groups was used: One group had to perform the task while being repeatedly interrupted by closing shutter glasses (occlusion condition), the other group had to perform the task without being interrupted (non-occlusion condition). As dependent variables we measured the number of correct answers to the questions concerning the names of the motorways and the text content. Each participant was randomly assigned to one of the groups. There were 13 participants in each group.

3.2. Results

Fig. 5 illustrates the results from the different conditions. It can be seen that the name of the motorway which was presented first was recalled correctly with an overall relative frequency of 0.62 ($SD=0.25$). Under the occlusion condition (with interruption) the relative frequency of correct answers was $M=0.38$ ($SD=0.08$), while under the non-occlusion condition (without interruption) the correct answer was given with a mean relative frequency of 0.85 ($SD=0.09$). A similar result was found concerning the question for the name of the second motorway in the sentences. Overall the relative frequency for a correct answer was 0.63 ($SD=0.31$). The occlusion condition led to a relative frequency of 0.33 ($SD=0.05$) for correct answers, while under the non-occlusion condition the relative frequency was 0.92 ($SD=0.06$). A two-factorial

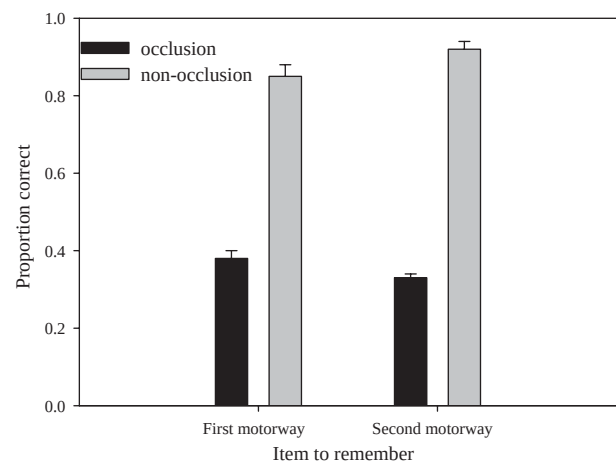


Fig. 5. Proportion of correctly recalled names of motorways under the occlusion and non-occlusion conditions.

ANOVA with the between-subjects factor occlusion condition (with or without occlusion) and the within-subjects factor motorway (first or second one) confirmed the significance of the difference in the relative frequency of correct answers between occlusion and non-occlusion condition, $F(1, 24) = 513.07$, $p < 0.001$. The interaction between occlusion condition and motorway was also significant, $F(1, 24) = 14.91$, $p = 0.001$. This was the case because a primacy effect was seen with occlusion, whereas without occlusion a recency effect was seen.

The results of the content questions are shown in Fig. 6. Overall the relative frequency for correct answers to the first question was 0.58 (SD=0.27). Under the occlusion condition the relative frequency of correct answers was 0.33 (SD=0.11). Under the non-occlusion condition the relative frequency of correct answers was 0.83 (SD=0.05). A similar result was found for the second question. The overall relative frequency of correct answers to the second question was 0.46 (SD=0.22). Under the occlusion condition the correct answer was given with a relative frequency of 0.26 (SD=0.09), while under the non-occlusion condition the correct answer was given with a relative frequency of 0.66 (SD=0.08). A two-factorial ANOVA with the between-subjects factor occlusion condition (with or without occlusion) and the within-subjects factor question (first or second one) revealed that the difference between conditions with and without occlusion was highly significant, $F(1, 24) = 265.9$, $p < 0.001$. Also the difference between the proportion of correct answers to the first and second question was highly significant, $F(1, 24) = 46.2$, $p < 0.001$. The interaction between the occlusion condition and the question factor was also significant, $F(1, 24) = 8.8$, $p = 0.007$. This was due to a greater difference in the proportion of correct answers between the first and second question for the condition without occlusion compared to the condition with occlusion.

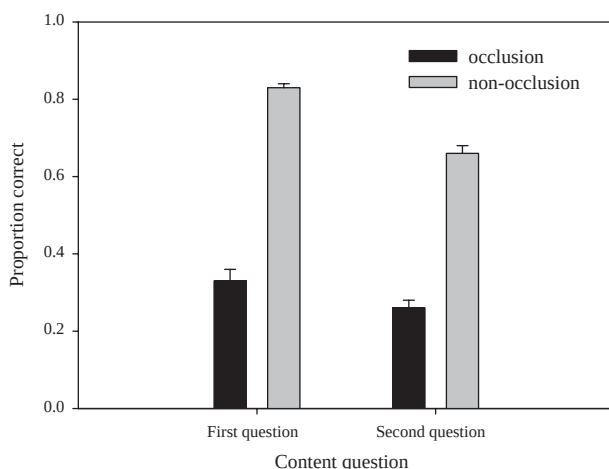


Fig. 6. Proportion of correctly answered content questions under the occlusion and non-occlusion conditions.

3.3. Discussion

Tijerina et al. (2000) strongly question the validity of the 15-second rule as a method to decide whether a task should be accessible while driving or not. They showed that this method cannot differentiate between tasks which can be performed while driving and those that cannot. One of their major criticisms was that the time needed to complete a task when it is just the task alone that is performed (total task time) does not take into account how the task is chunked or partitioned into smaller units when performed while driving. But this is essential for the evaluation of the implications the task has on driver safety, as each chunk's completion should not take more than 1–2 s. Otherwise this would force the driver to direct his attention away from the road for too long a time leading to a deterioration of driving performance (Zwahlen et al., 1988). Our experiment demonstrates this point. The task of this experiment was designed to follow a static criterion (total task time around 15 s) but performance under the occlusion condition showed a significant drop in quality when participants were interrupted during task completion. Only about one third of the trials were answered correctly under the occlusion condition. This holds true both for questions requiring specific pieces of information presented in the text (names of motorways) and for questions on the general content of the text. Therefore, if this task was undertaken while driving it could lead either to a deterioration of driving performance, as the driver has to attend to the display for too long a time period, or to a situation where the task could not be completed satisfactorily. The reason for this lies in the dynamic structure of the task which makes it very difficult to chunk the task into smaller units. Information presented during the interruption phase when the display was not visible, was lost and could not be recovered. The task used in this experiment may seem to be an extreme example, but it nevertheless demonstrates the principle of the argument. The total task time alone cannot be used as the only criterion to evaluate whether the design of the interface allows a task to be accessed while driving, because it neglects how the task is partitioned during task performance. By means of the occlusion technique it seems to be more adequate to separate tasks that are chunkable without a substantial loss of performance from tasks that are not.

4. General discussion—practical consequences

Results from previous laboratory experiments suggest that the occlusion technique might be an appropriate instrument for evaluating in-car information systems

with regard to two critical aspects. First, the experiments showed that the occlusion technique reliably differentiates between displays of different visual complexity (Krems et al., 2000). Secondly, the technique was useful for assessing how easily a dialogue could be resumed after an interruption. The results also showed that it is possible to measure the influence of dialogue interaction on tasks that have to be performed concurrently (Keinath et al., 2001). This seems especially interesting as it offers the chance of assessing the level of impairment the interaction with an in-car information system can have on driving. The two experiments described in the current paper supplement these findings by adding results from a driving study and from a comparison to an alternative evaluation procedure.

In the first study, the same task and material was used under three different conditions: static, occlusion, and driving. The pattern of results shows two important aspects of the occlusion technique as a method for evaluating in-car information systems. The occlusion technique on its own does not simulate all the demands of driving a car in a realistic environment. That is, driving as a primary task impairs interaction with an in-car information system more than the occlusion technique does. This is reflected by longer total task times and higher error rates under the driving condition than under the occlusion condition. However, the occlusion technique influences subject performance in a way comparable to driving a car. This is indicated by the similar influence of input length under both the occlusion and driving conditions. The error rates for the two most frequent error categories also reflect the greater similarity between the driving and the occlusion conditions. They did not differ in the driving and occlusion conditions, but they did differ in the driving and the non-occlusion conditions.

The fact that this technique is less demanding than a real traffic situation is not necessarily a disadvantage. Earlier results indicate that the cognitive load can easily be increased by adding an additional task to the occlusion condition (see Keinath et al., 2001). Therefore, the occlusion technique allows the experimenter to simulate different traffic situations by using additional tasks of different complexity. Our results indicate that functions which do not pass an occlusion test, which is less demanding than real traffic, should not be considered suitable for use while driving. In general, the occlusion technique seems to be a reliable method, and one which can be expanded upon (for example, by adding secondary tasks), to evaluate in-car information systems that involve visual displays.

The second experiment compared a global criterion (total task time), a very easy procedure proposed for deciding if a task is executable while driving without violating safety restrictions, with the occlusion techni-

que. The results showed that tasks that are definitely not suitable under traffic conditions, because they fail the chunkability criterion, may comply with the 15-second rule. However, by means of the occlusion technique this type of tasks can be identified.

To summarise, results of both experiments show that the occlusion technique is a useful and viable method for evaluating in-car information systems. This technique allows to assess whether presented pieces of information can be interpreted within a tolerable time scale. Furthermore, it is possible to systematically control how long information, e.g. presented on a display, is available to the driver (presentation time) and is hidden from sight (occlusion time). The accuracy with which the presented pieces of information are interpreted can be used to evaluate the information system. Additionally the point of task interruption can be controlled. Therefore, the occlusion technique provides means of assessing whether a task can be easily interrupted.

Further investigations should answer the question of whether the performance under occlusion conditions can be made more equivalent to the performance under real traffic conditions by adding additional tasks to fill the occlusion phase or by lengthening the occlusion phase. The effect of the length of the occlusion phase on task performance is not sufficiently examined by now. Another issue of future research should be to use the occlusion method in order to assess what proportion of a single task can be carried out without visual control.

Acknowledgements

This study was supported by BMW Group, Munich.

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